

Identifiability in observational data and in group factor analysis

When labels are needed, when multi-view structure substitutes for them

Question

Are there identifiability issues in observational data that get resolved only when we also have a label per data point? Is this also true for the group factor analysis (GFA) setting?

Mostly yes — but the multi-view setting changes the picture meaningfully.

1. The single-view story

For nonlinear representation learning on a single view of i.i.d. observational data, **the nonlinear ICA model is fundamentally non-identifiable** (Hyvärinen & Pajunen 1999). For any nonlinear mixing f producing the data, infinitely many other nonlinear mixings give the same observed distribution. The standard fix is conditioning on an auxiliary variable u :

$$p(\mathbf{z} \mid u) = \mathcal{N}(\mu(u), \Sigma(u)),$$

which Hyvärinen & Morioka (2016, “TCL”; using a time index u) and Khemakhem et al. (2020, “iVAE”; any auxiliary) showed restores identifiability up to a *component-wise* transform. The label, time index, environment, or class breaks the rotational/mixing ambiguity.

So in a single-view nonlinear ICA setting your statement is correct: per-sample auxiliary information is essentially required.

2. What changes in GFA / multi-view

Multi-view structure is itself an identifiability-restoring signal — you don’t necessarily need per-sample labels.

2.1 Linear GFA (Klami 2014; MOFA, MuVI)

Identifiable up to factor permutation/sign **without labels**, via two combined mechanisms:

1. **Sparsity priors on per-view loadings** (ARD / spike-and-slab) — break the rotation ambiguity that plagues vanilla factor analysis. With column-sparse W_v , almost every rotation destroys sparsity, so the maximum-a-posteriori solution is essentially unique up to permutation/sign.
2. **The multi-view structure itself** — different sharing patterns across views constrain which factors can be active where. A rotation that mixes a shared and a view-private factor is penalized by the per-view ARD.

Linear GFA never required per-sample labels. Identifiability comes from the priors plus the view structure.

2.2 Nonlinear / deep multi-view (Gresele 2020; Daunhawer 2023)

Two formal results that hold *without* per-sample labels:

- **Gresele, Locatello, Schölkopf 2020 — “Multi-view Nonlinear ICA”**. Two views with shared content $\mathbf{z}^c$ and view-private style $\mathbf{z}_v^p$ are sufficient to identify $\mathbf{z}^c$ up to a component-wise transform, under smoothness

+ independence assumptions on the data-generating process. **Multiple views act as the auxiliary signal.**

- **Daunhawer, Bauer, Locatello et al. (ICLR 2023)** — “**Identifiability Results for Multimodal Contrastive Learning**”. Multimodal contrastive learning (CLIP-style) provably recovers $\mathbf{z}^c$ under analogous assumptions. Same lesson: views serve as auxiliaries to each other.

So for the *shared* content in a multi-view setup, identifiability holds without labels. For the *private* content per view, the situation is weaker — you typically need additional structure (sparsity, independence, monotonicity, sign constraints) or you fall back on labels.

3. Where labels still help even in multi-view

Three places per-sample labels strengthen identifiability beyond what multi-view alone gives:

1. **Recovering private factors** that aren’t pinned down by the multi-view assumption.
2. **Going from “up to component-wise transform” to “up to permutation/sign”** — a tighter form of identifiability that’s easier to interpret biologically.
3. **Causal / counterfactual claims** — iVAE-family results give this only with auxiliaries.

4. Implications for the chemogenetic-screens setup

The drug $\times$ pert $\times$ gene-logFC data has a **doubly-identifying structure**:

- **Multi-view**: 16 drug contexts as views.
- **Per-sample auxiliary**: each cell carries a **target_gene** label (the perturbation), giving $\approx 2\{, \}400$ distinct classes. In iVAE language, this is exactly the auxiliary that stratifies the latent prior.

Identifiability-wise this is unusually favorable. Concretely:

- **For linear GFA** (e.g. MOFA+ on the per-drug PosteriorMean matrices): rely on per-view ARD + view structure. No need for labels in the model — they’re implicit because each row is a different perturbation.
- **For deep multi-view models** (e.g. the 14_2 / 14_3 VAE notebooks): condition the encoder on **target_gene** to put yourself fully in the iVAE setting — that delivers component-wise identifiability for *both* shared and private.
- **Cell-level OLS on `sc.tl.score_genes` outputs**: the per-cell **target_gene** label is exactly the auxiliary, and the joint OLS estimates per-perturbation coefficients β_p that are well-defined under this setup.

5. Summary table

Setting	Needs per-sample labels?	Identifiability granularity
Single-view nonlinear ICA (vanilla VAE on observational data)	Yes — the only fix	Up to component-wise transform (with auxiliary)
Linear GFA (Klami, MOFA, MuVI)	No — ARD + multi-view structure suffices	Up to permutation / sign
Multi-view nonlinear ICA , <i>shared content</i> (Gresele 2020)	No — views substitute for labels	Up to component-wise transform
Multi-view nonlinear ICA , <i>private content</i>	Often weak without extra assumptions	Need sparsity / independence priors or labels
Multi-view + per-sample labels (iVAE-flavour)	Both signals available	Tightest identifiability

6. References

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